

Dorsa Dolatkhah Ouchbolagh

Polymer Engineer — M.Sc. Candidate, University of Tehran

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EDUCATION

M.Sc. in Polymer Engineering

2026 – Present

School of Polymer Engineering, Department of Chemical Engineering, College of Engineering, University of Tehran

B.Sc. in Polymer Engineering

2022 – 2026

School of Polymer Engineering, Department of Chemical Engineering, College of Engineering, University of Tehran

GPA: 3.60 / 4.00 (18.01 / 20)

High School Diploma, Mathematics and Physics

2018 – 2021

Nikan Public High School, Tehran, Iran

GPA: 4.00 / 4.00 (19.45 / 20)

RESEARCH INTERESTS

Sustainable & Biodegradable Polymers · AI and Machine Learning in Polymer Science · Polymer Recycling · Smart and Stimuli-Responsive Biosensors · Biomedical Polymers · Bio-based Polymer Composites · Bio-inspired materials

SELECTED COURSEWORK

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|--|---|
| • Rubber and Elastomer Engineering 18.0/20 | • Polymer Rheology 18.6/20 |
| • Physical and Mechanical Properties of Polymers 19.9/20 | • Physical Chemistry of Polymers 20.0/20 |
| • Synthesis of Polymers 18.5/20 | • Resins Engineering 19.6/20 |
| • Polymers and the Environment 20.0/20 | • Natural Polymers 20.0/20 |
| • Polymer Molecular Analysis 19.5/20 | • Principles of Polymerization <i>in progress</i> |

RESEARCH and PROJECTS

- Bachelor's thesis: a review of bio-based phase change materials for thermal and energy management, with applications in building structures, batteries, and personal thermal-management patches.
- Designed and Fabricated bagasse-based food containers embedded with phase change materials to control heat dissipation and cold storage.
- A Class project on smart, sustainable and biodegradable polymers (e.g., PLA) using 4D filament printing with heat sensitivity for shape memory and reversibility properties.

PUBLICATION

- A Review on Biomass Selection for Phase Change Material-Embedded Composites: Criteria, Material Properties, and Design Considerations. Manuscript in Progress

EXPERIENCE

Teaching Assistant — Calculus I & II

Fall 2022

Department of Chemical Engineering, University of Tehran

- Assisted the course instructor in grading assignments and supporting classroom activities.

Teaching Assistant — Differential Equations (ODE)

Spring 2023

Department of Chemical Engineering, University of Tehran · Course instructor: Prof. H. Rahami

- Guided undergraduate students during tutorials and problem-solving sessions.
- Communicated complex engineering concepts in a clear and accessible manner.

Laboratory Intern

Summer 2025

Rubber Research and Engineering Company (VardAvard), Tehran, Iran · 2-month internship

- Contributed to interdisciplinary research projects supporting rubber and automotive industry applications.

- Senior Assistance in Polymer processing laboratory and rheological, physical and mechanical test procedures to ensure accurate experimental procedures and properties based on ASTM standards.

SKILLS

Software	MATLAB, Python, Aspen Plus
Technical Analysis	Rheometry, DSC, XRD, TGA, DMTA

AWARDS AND HONORS

- Ranked 1st among 23 undergraduates in the B.Sc. Polymer Engineering program, School of Polymer Engineering, University of Tehran.
- Admitted to M.Sc. programs at the University of Tehran, Amirkabir University of Technology, and Sharif University of Technology without entrance examination, on the basis of academic excellence.

LANGUAGE SKILLS

Persian	Native
English	Advanced — TOEFL iBT, Spring 2026: [score pending]; Certificate of Proficiency in English, Bloomsbury International (UK), 2018: 89/100

REFERENCES

Dr. SeyedMostafa Hosseini — Assistant Professor, Department of Chemical Engineering, School of Polymer Engineering, University of Tehran

Email: smostaffahosseini@ut.ac.ir

Dr. SeyedHassan Jafari — Professor, Department of Chemical Engineering, School of Polymer Engineering, University of Tehran

Email: shjafari@ut.ac.ir